Varendra University

Dept. of Computer Science & Engineering (CSE)

Microprocessor

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

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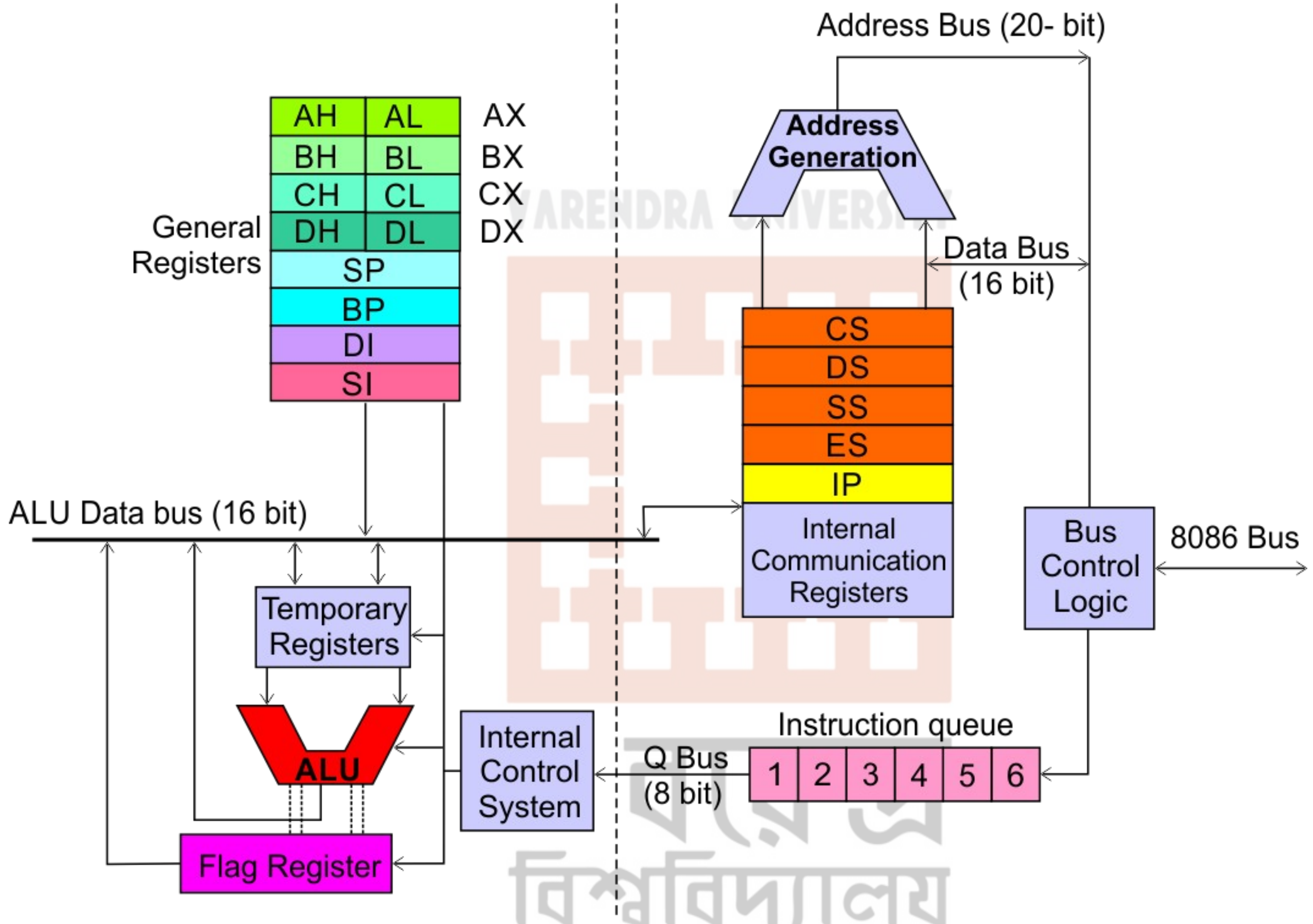
Outline

1. 8086 Microprocessor Organization.
2. Functional Block of 8086 Microprocessor.
3. How does queue speed up the processing of the 8086 microprocessor?
4. Register of 8086 Microprocessor.
5. Instruction set of 8086 microprocessor.

8086 Microprocessor Organization



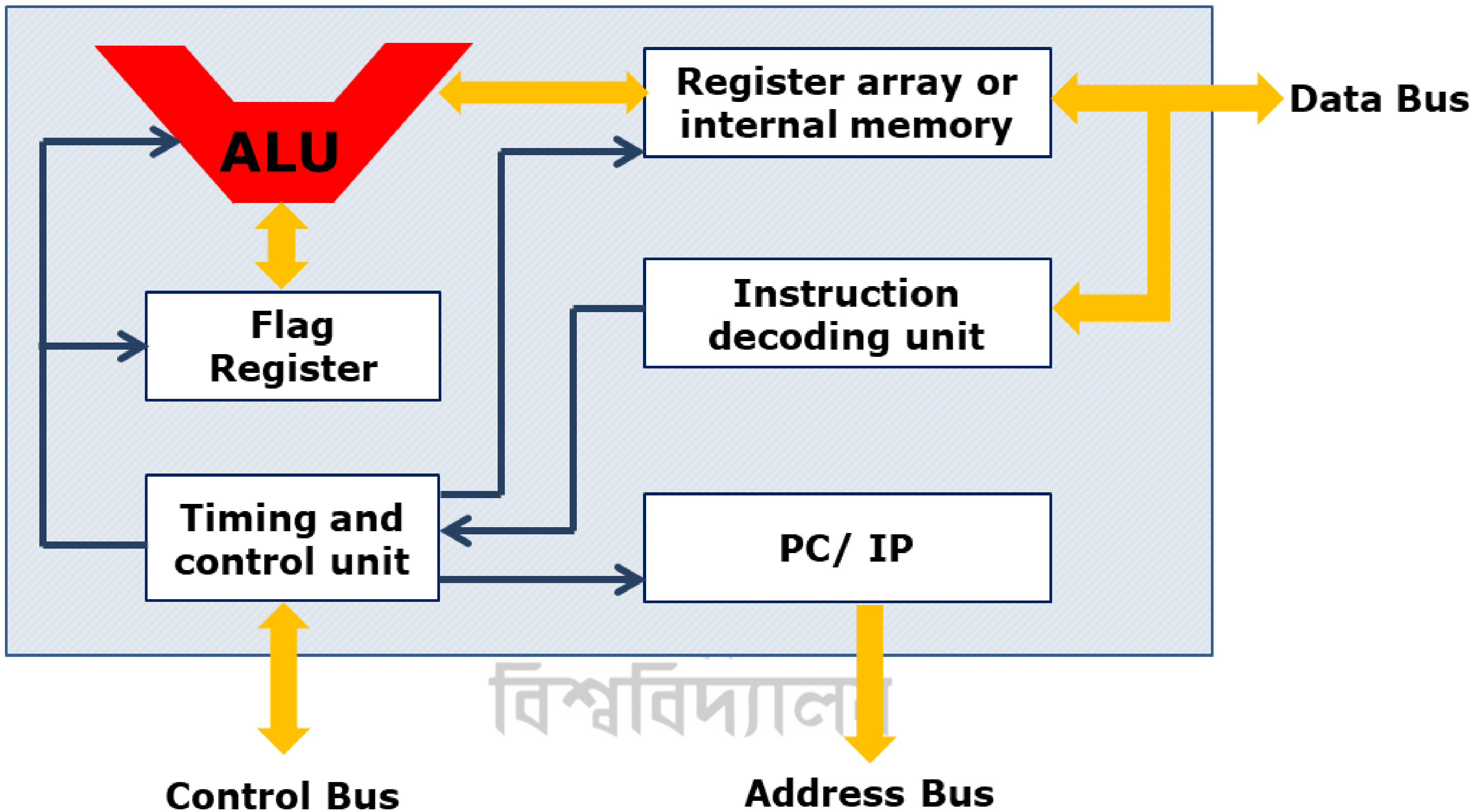
Execution Unit (EU)

- **Instruction decode and execute.**
- **Arithmetic and logical operation.**

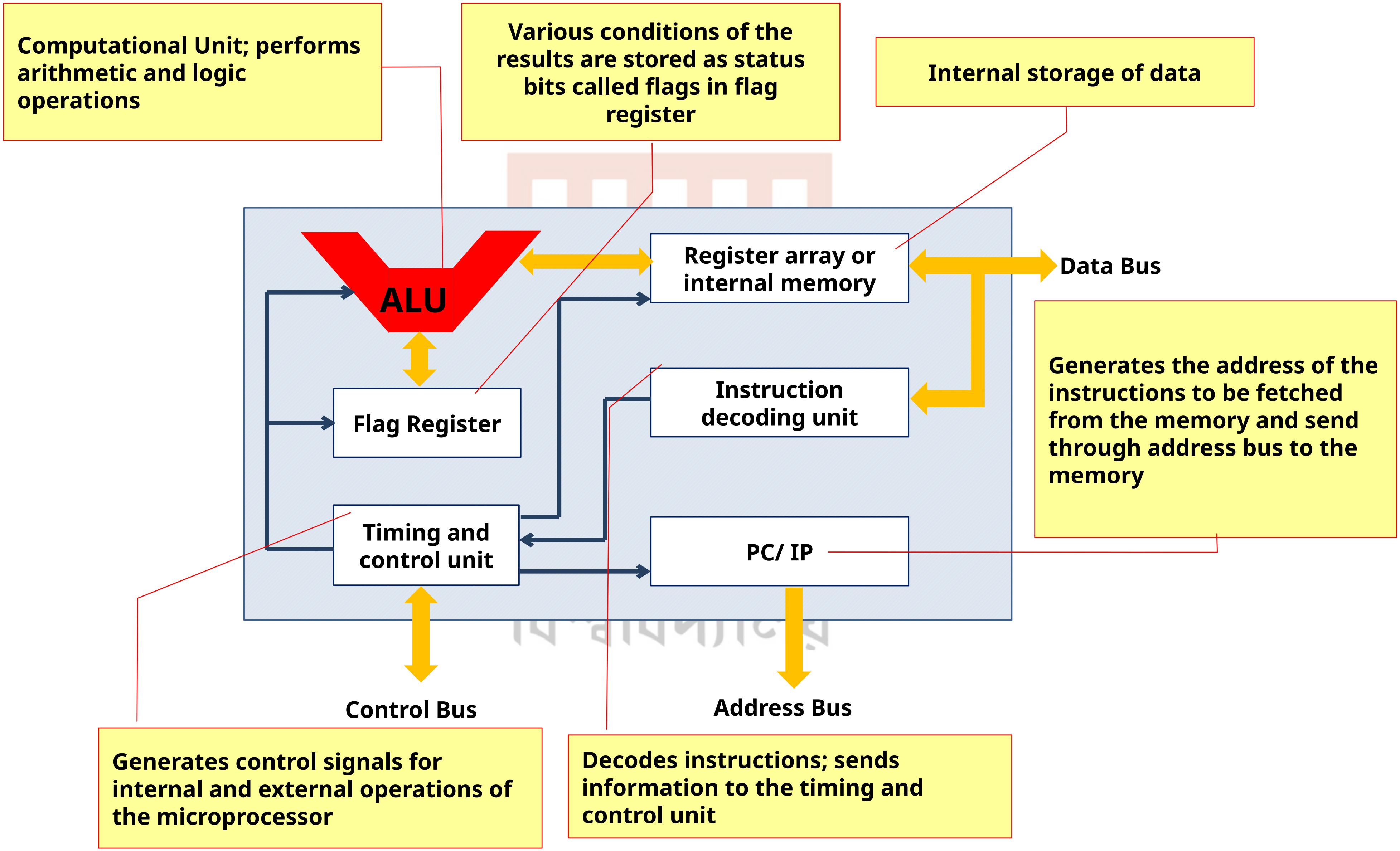
Bus Interface Unit (BIU)

- **Instruction fetch.**
- **Address generation.**
- **Memory and I/O communication.**

Functional Block of 8086 Microprocessor



Functional Block of 8086 Microprocessor



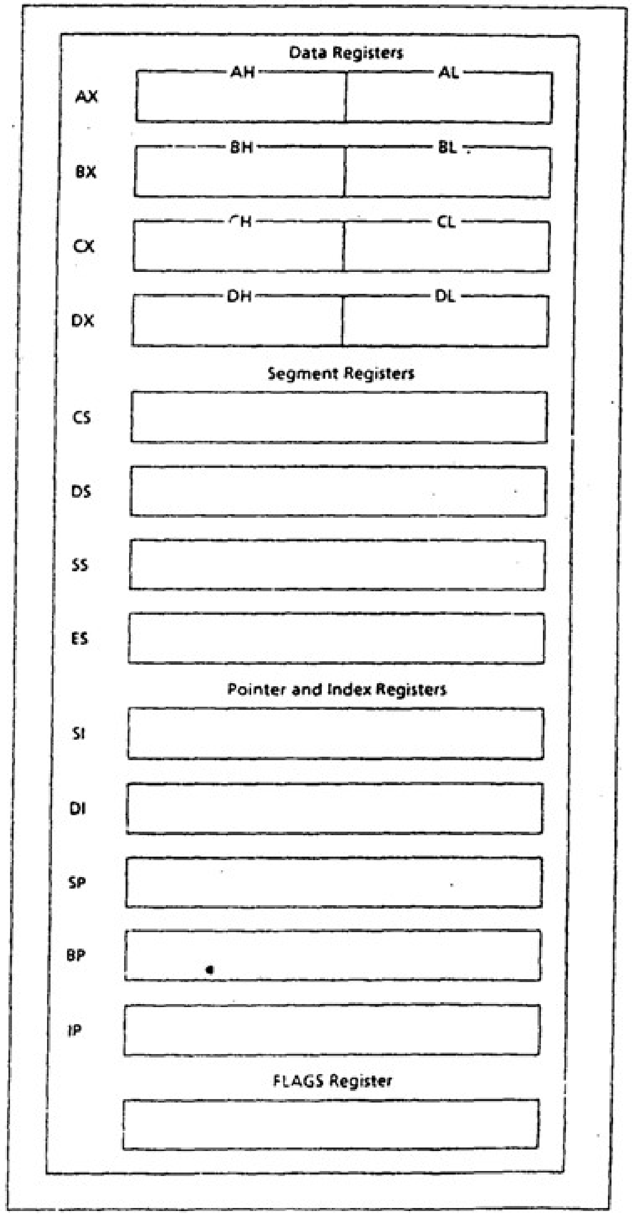
How does queue speed up the processing of the 8086 microprocessor?

The 8086 microprocessor's 6-byte instruction queue increases speed through instruction prefetching. The BIU fetches instructions and stores them in the queue. At the same time, the EU executes the instructions. Because of instruction fetching and execution overlap, the processor's idle time is reduced and execution speed increases.

Register of 8086 Microprocessor

Category	Bits	Register Name
Data / General Purpose Registers	16	AX, BX, CX, DX
	8	AH, AL; BH, BL; CH, CL; DH, DL
Pointer	16	SP (Stack Pointer), BP (Base Pointer)
Index	16	SI (Source Index), DI (Destination Index)
Segment	16	CS (Code Segment) DS (Data Segment) SS (Stack Segment) ES (Extra Segment)
Instruction	16	IP (Instruction Pointer)
Flag	16	FR (Flag Register)

Register of 8086 Microprocessor



Register of 8086 Microprocessor

Data Register: Manipulates data.

- **AX (Accumulator Register):** Used in arithmetic, logic and data transfer instructions because it generates the shortest machine code.
- **BX (Base Register):** It is mainly address register. It contains data pointer which is base index. It indirectly addresses the register.
- **CX (Count Register):**
  - Program loop construction.
  - String manipulation.
  - Acts as a counter in shift or rotate instruction.
- **DX (Data Register):**
  - Works as the port number during the input and output operation.
  - Used in multiplication and division.

Register of 8086 Microprocessor

Segment Register: Stores address of instructions and data.

- **CS (Code Segment):** Used for storing instructions.
- **DS (Data Segment):** Used for storing data.
- **SS (Stack Segment):** Used as stack and store returning address.
- **ES (Extra Segment):** Used for storing destination data.

Register of 8086 Microprocessor

Pointer and Index Register: Points to the memory location. It is used in arithmetic and other operations.

- **SP (Stack Pointer):** Used for accessing the stack segment (SS).
- **BP (Base Pointer):** Used primarily to access data on the stack.
- **SI (Source Index):** Used to point memory locations in the data segment addressed by DS.
- **DI (Destination Index):** Performs the same functions as SI.

Register of 8086 Microprocessor

Flag Register: To indicate the status of the microprocessor. It contains the status and control flag.

- **Status flag:** Reflects the result of an instruction executed by the CPU.
- **Control flag:** Used to enable or disable certain operation of microprocessor.

Instruction set of 8086 microprocessor.

Data Transfer Instructions:

Instruction	Description
MOV	Transfer data between registers or between register and memory.
PUSH	Push register/memory data onto the stack.
POP	Pop data from stack into register/memory.
XCHG	Exchange data between two registers/memory.
IN	Input from a port to register.
OUT	Output from register to a port.
LEA	Load effective address of memory into a register.

Instruction set of 8086 microprocessor.

Arithmetic Instructions:

Instruction	Description
ADD	Add two values.
ADC	Add with carry.
SUB	Subtract one value from another.
SBB	Subtract with borrow.
MUL	Unsigned multiplication.
IMUL	Signed multiplication.
DIV	Unsigned division.
IDIV	Signed division.
INC	Increment value by 1.
DEC	Decrement value by 1.
NEG	Negate (2's complement).
CMP	Compare two values (subtract but don't store

Instruction set of 8086 microprocessor.

Logical Instructions:

Instruction	Description
AND	Logical AND.
OR	Logical OR.
XOR	Logical Exclusive OR.
NOT	Logical Complement.
TEST	Logical AND (Result not stored).

Instruction set of 8086 microprocessor.

Shift and Rotate Instructions:

Instruction	Description
SHL	Shift bits left.
SHR	Shift bits right.
ROL	Rotate bits left.
ROR	Rotate bits right.
RCL	Rotate through carry left.
RCR	Rotate through carry right.

Instruction set of 8086 microprocessor.

Branching Instructions (Control Transfer

Instruction	Description
JMP	Unconditional jump.
RET	Return from a subroutine.
JE / JZ	Jump if equal (zero flag set).
JNE / JNZ	Jump if not equal (zero flag clear).
JG	Jump if greater.
JL	Jump if less.
JGE	Jump if greater or equal.
JLE	Jump if less or equal.
LOOP	Loop with CX register.
INT	Interrupt execution.

Instruction set of 8086 microprocessor.

String Manipulation Instructions:

Instruction	Description
MOVS	Move string from one location to another.
LODS	Load string element into AL/AX.
STOS	Store string element from AL/AX.
SCAS	Scan a string for a value.
CMPS	Compare two strings.
REP	Repeat instruction while $CX \neq 0$.

Instruction set of 8086 microprocessor.

Processor Control Instructions:

Instruction	Description
STC	Set Carry flag.
CLC	Clear Carry flag.
CMC	Complement Carry flag.
STD	Set Direction flag.
CLD	Clear Direction flag.
STI	Set Interrupt flag (enable interrupts).
CLI	Clear Interrupt flag (disable interrupts).
HLT	Halt processor execution.
NOP	No operation (does nothing).
WAIT	Wait for processor signal.
ESC	Escape (for co-processor operations).
LOCK	Lock the bus to prevent other processors from

Instruction set of 8086 microprocessor.

The 8086 **instruction set** consists of:

- **Data Transfer Instructions:** Move data.
- **Arithmetic Instructions:** Perform calculations.
- **Logical Instructions:** Bitwise operations.
- **Shift and Rotate Instructions:** Bit manipulation.
- **Branching Instructions:** Control program flow.
- **String Instructions:** Process strings.
- **Processor Control Instructions:** Control CPU operations.